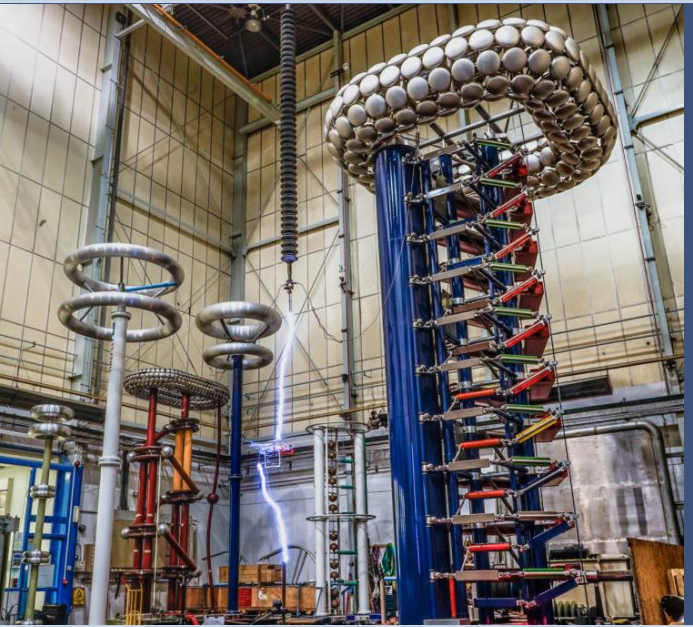


# EE4047

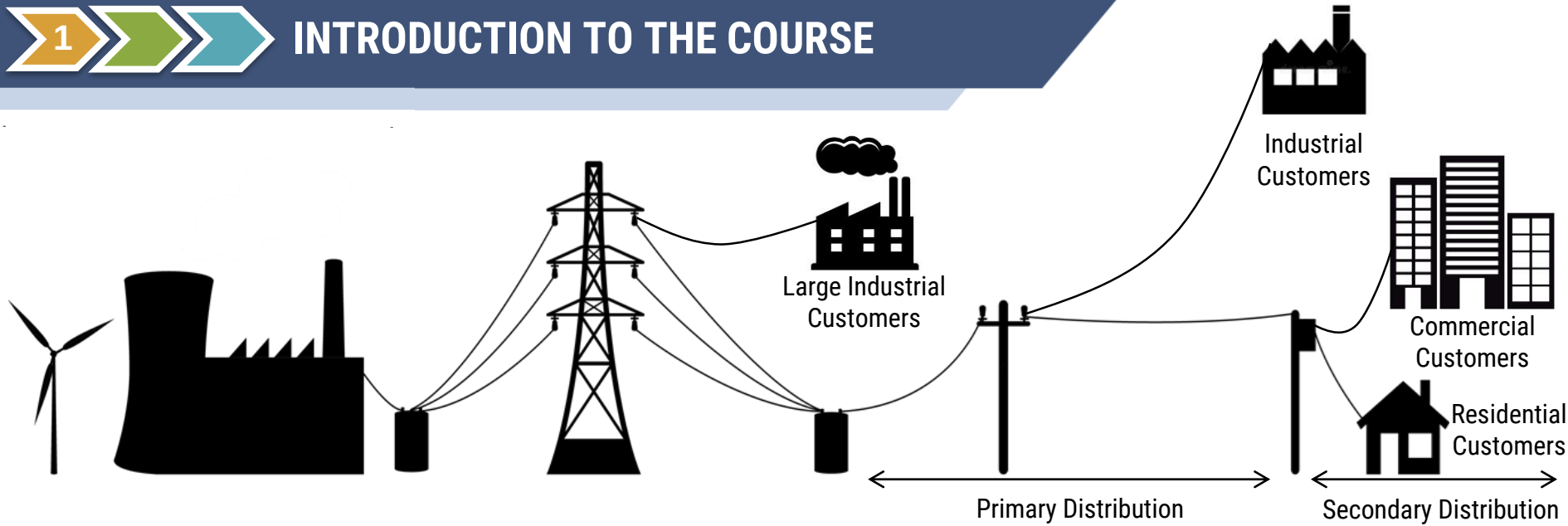
## High Voltage Techniques I

**Week 2.1 – Monday Lesson**

**Asst. Prof. Dr. M. Alparslan Zehir**

**06.03.2023**

# INTRODUCTION TO THE COURSE




**Generation**

(Centralized Power plants located near available resources)

10-30 kV

**Transmission**

(Transmit energy over long distances to load centers, using overhead lines and **high voltage**)

66-750 kV AC or
500-1500 kV DC

**Distribution**

(Transmit energy to its final destination using overhead lines, underground cables, using medium voltage and low voltage)
0.4 to 36 kV

**Retail**

(End-uses of customers such as lighting, appliances, heating etc.)

# 2 TURKISH NATIONAL ELECTRIC POWER SYSTEM

## Brief History of Turkish National Electric Power System

- The first power generation in Turkey had been from a 2 kW dynamo connected to water-mill in Tarsus-Mersin, in 1902.

- In 1914, coal-fired Silahtaraga Power Plant (3*6 MW) was put into service in Istanbul.







# 2 TURKISH NATIONAL ELECTRIC POWER SYSTEM

## Brief History of Turkish National Electric Power System

- Operation of the power industry under the leadership of State was started upon the establishment of Etibank in 1935.

- Turkish Electricity Authority (TEK) was founded in1970 aiming at the rendering of generation, transmission, distribution and trade of electricity services needed by the country as per the general energy and economy policies of the State.

- TEK was later (1993) split into two separate state economic enterprises namely:

  Turkish Electricity Generation Transmission Co. (TEAŞ): Responsible from power generation and transmission

  Turkish Electricity Distribution Co. (TEDAŞ)

# 2 TURKISH NATIONAL ELECTRIC POWER SYSTEM

## Brief History of Turkish National Electric Power System

- In 2001, the Government implement "Program for Economic Stability and Fighting Against Inflation", main tasks of which include;

  Restructuring of the power sector,

  Transition into a free electricity market ensuring free competition,

  Setting up of separate companies for generation, transmission, wholesale and distribution of electricity and

  Privatization of the all public entities other than that of transmission.

# TURKISH NATIONAL ELECTRIC POWER SYSTEM

## Brief History of Turkish National Electric Power System

- In 2001, TEAŞ has been restructured in three separate state economic enterprises:

  Electricity Generation Co. (EÜAŞ) partly privatized

  Turkish Electricity Transmission Co. (TEİAŞ) and

  Turkish Electricity Contracting and Trading Co. (TETAŞ) (closed in 2018)

- In 2006, TEİAŞ has opened PMUM for wholesale electricity market operations, which transformed later into EPİAŞ in 2015.

- Turkish Electricity Distribution Co. (TEDAŞ) partly privatized (21 regions)



# TURKISH NATIONAL ELECTRIC POWER SYSTEM

## Brief History of Turkish National Electric Power System

- In 2015 TEİAŞ has become an observer member of ENTSO-E (European Network of Transmission System Operators for Electricity) and Turkish National Electric Power System has joined the large interconnected network of national grids consisting of 35 countries and 42 Transmission System Operators (TSO).



## Turkish National Electric Power System Today

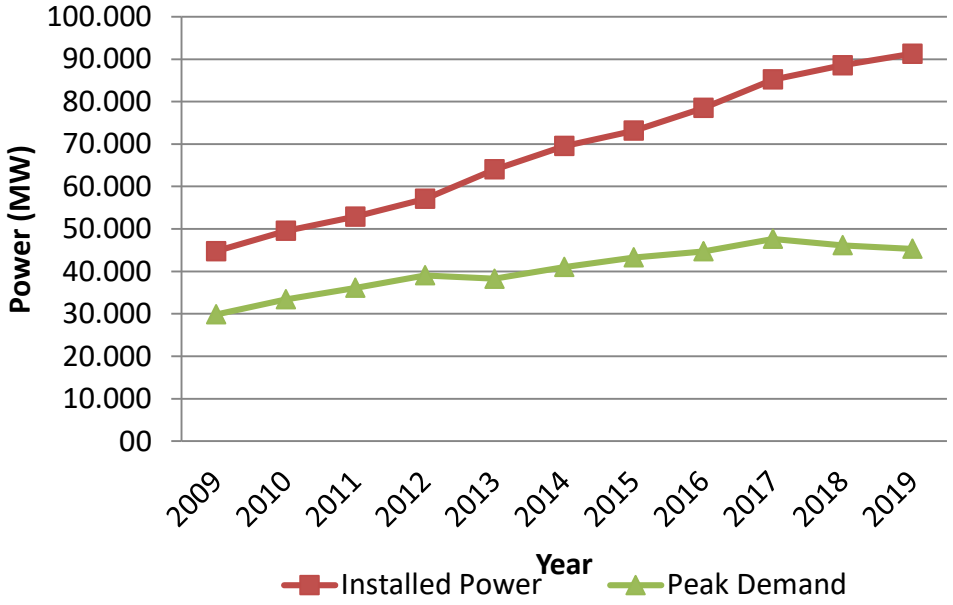
- Annual change of peak demand and overall energy consumption

| Year | Peak Demand (MW) | Change (%) | Energy Consumed | Change (%) | Minimum Demand | Min. demand / peak demand (%) |
|---|---|---|---|---|---|---|
| 2010 | 33.392 | 11,8 | 210.434 | 8,4 | 13.513 | 40 |
| 2011 | 36.122 | 8,2 | 230.306 | 9,4 | 14.822 | 41 |
| 2012 | 39.045 | 8,1 | 242.370 | 5,2 | 13.922 | 36 |
| 2013 | 38.274 | -2,0 | 246.357 | 1,6 | 14.800 | 39 |
| 2014 | 41.003 | 7,1 | 257.220 | 4,4 | 14.927 | 36 |
| 2015 | 43.289 | 5,6 | 265.724 | 3,3 | 16.269 | 38 |
| 2016 | 44.734 | 3,3 | 279.286 | 5,1 | 17.448 | 39 |
| 2017 | 48.807 | 9,1 | 296.702 | 6,2 | 18.336 | 38 |
| 2018 | 49.304 | 1,0 | 304.167 | 2,5 | 18.212 | 37 |
| *2019 | 49.281 | -0,05 | 303.674 | -0,2 | 18.300 | 37 |

# TURKISH NATIONAL ELECTRIC POWER SYSTEM

## Turkish National Electric Power System Today

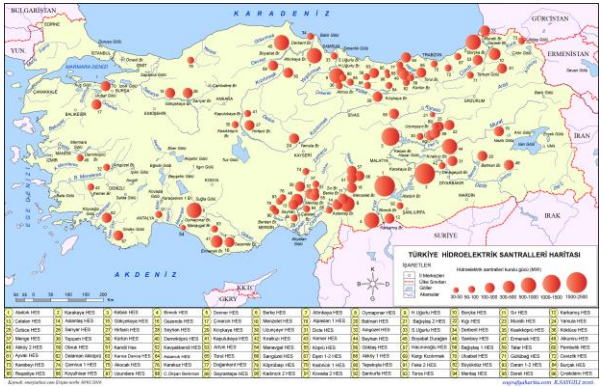
- Annual change of peak demand vs. Annual increase of installed power



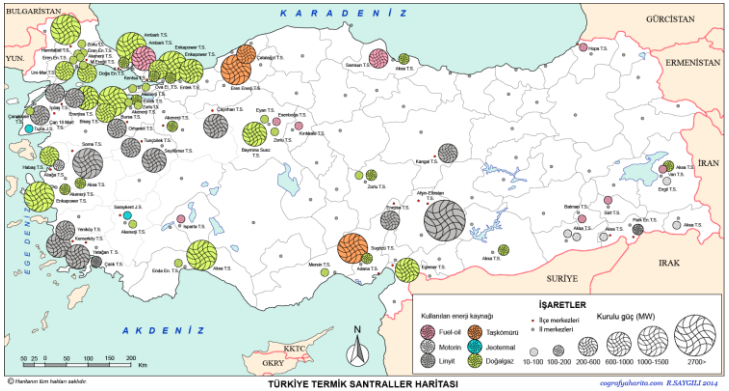
# TURKISH NATIONAL ELECTRIC POWER SYSTEM

## Turkish National Electric Power System Today

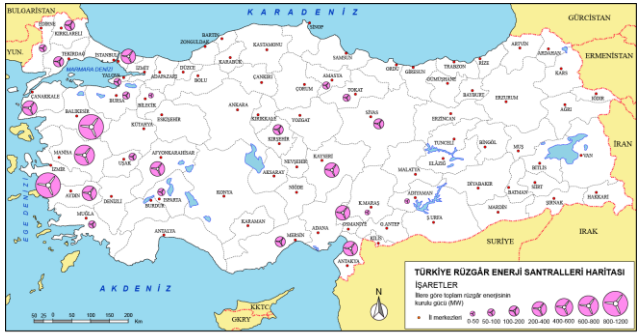
- Power plants




Hydropower Plants




Thermal Power Plants




Wind Farms

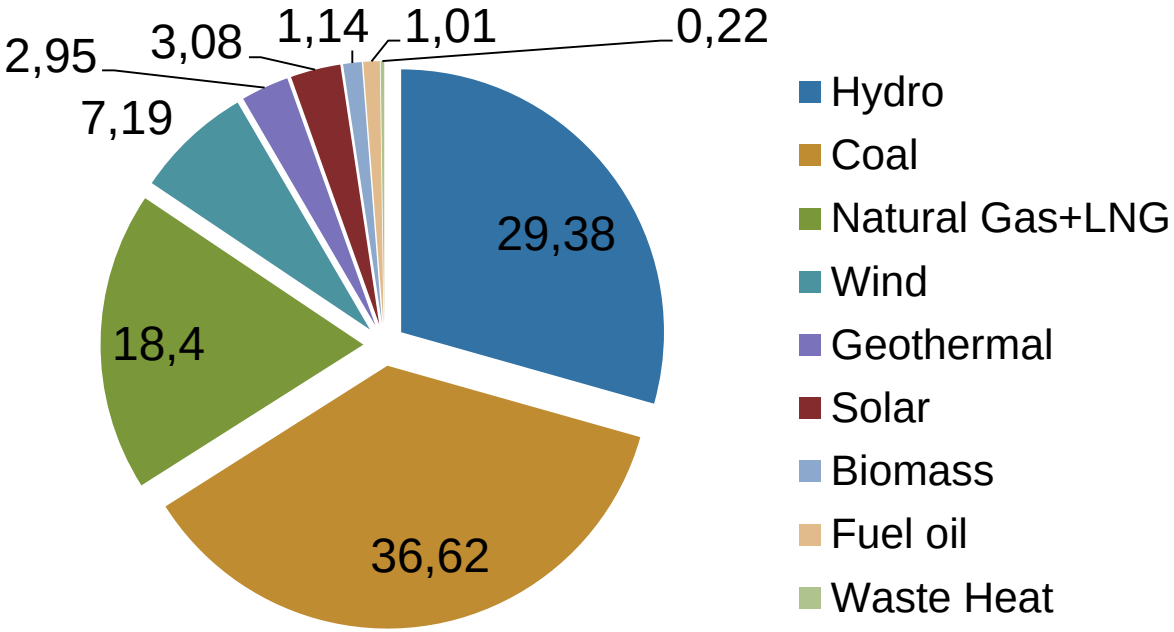



Geothermal Power Plants

# TURKISH NATIONAL ELECTRIC POWER SYSTEM

## Turkish National Electric Power System Today

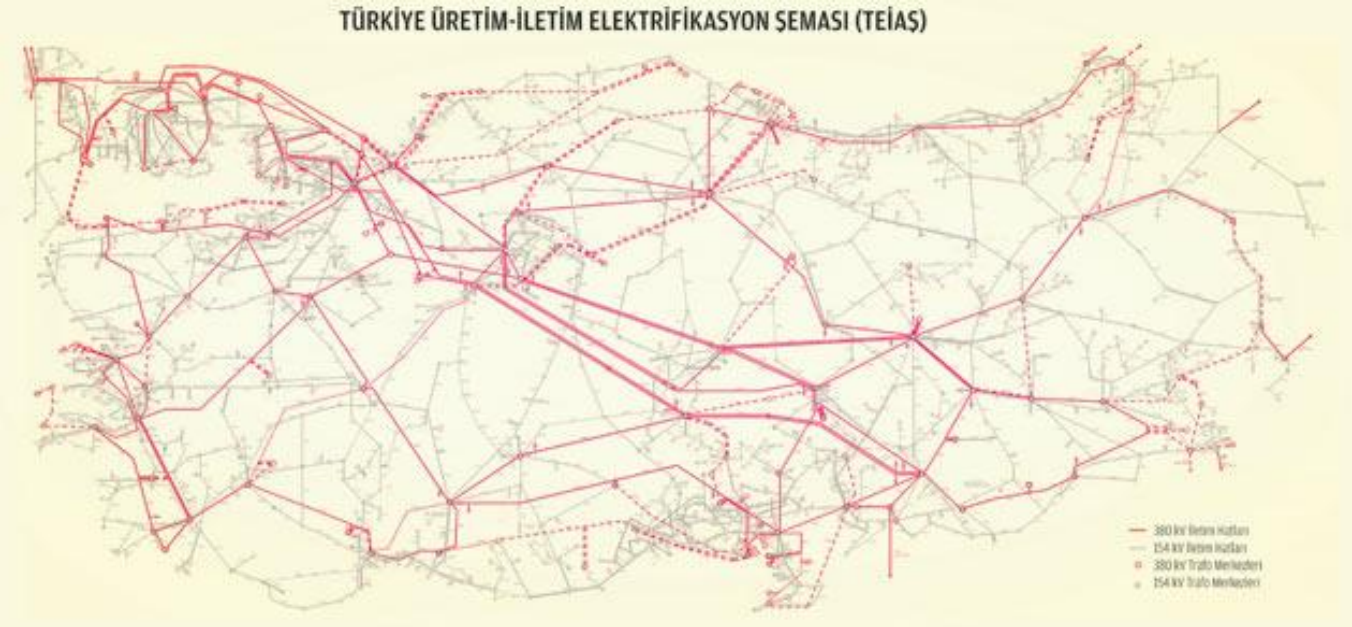
- Percentage share of electricity generation per source type



# TURKISH NATIONAL ELECTRIC POWER SYSTEM

## Turkish National Electric Power System Today

- Transmission lines




- 220 kV Interconnection lines connect Turkish system to Georgia, Bulgaria and Greece.

## Turkish National Electric Power System Today

- Annual increase of the overall length of overhead transmission lines

| Year | 400 kV | 220 kV | 154 kV | 66 kV | Total |
|---|---|---|---|---|---|
| 2010 | 15.734 | 85 | 32.906 | 509 | 49.234 |
| 2011 | 15.978 | 85 | 32.878 | 509 | 49.451 |
| 2012 | 16.344 | 85 | 33.481 | 509 | 50.418 |
| 2013 | 16.808 | 85 | 33.943 | 509 | 51.345 |
| 2014 | 17.683 | 85 | 35.132 | 509 | 53.409 |
| 2015 | 19.071 | 85 | 37.449 | 140 | 56.744 |
| 2016 | 21.029 | 85 | 38.682 | 139 | 59.934 |
| 2017 | 22.506 | 85 | 43.152 | 110 | 65.853 |
| 2018 | 23.159 | 85 | 44.369 | 110 | 67.723 |
| 2019 | 23.847 | 85 | 45.470 | 110 | 69.511 |

- As of 2019, 429,36 km 154 kV, 77,75 km 400kV underground cable, 15,96 km 400 kV submarine power cable

## Turkish National Electric Power System Today

- Annual increase of the overall length of distribution lines

| | 33 kV | 15.8 kV | 10.5 kV | 6.3 kV | Other | MV Total | 0.4 kV (LV) | Total |
|---|---|---|---|---|---|---|---|---|
| Overhead | 395.617,2 | 17.897,8 | 183,2 | 1.725,5 | 5.303,6 | 420.727,2 | 532.840,5 | 953.567,8 |
| Underground | 62.759,9 | 3.076,2 | 3.979,7 | 974,2 | 284,3 | 71.074,2 | 129.875,1 | 200.949,3 |
| Total | 458.377,1 | 20.974,0 | 4.162,9 | 2.699,6 | 5.587,9 | 491.801,5 | 662.715,6 | 1.154.517,1 |

# END OF THE LESSON

## Any questions?

Check out the Power Engineering playlist with 32 videos on Youtube:

https://www.youtube.com/watch?v=JwRTpWZReJk&list=PLLZI7WaEmZi3F134eDujF5ZNGxA1E0dkm